UniVerse

A Campus Companion

Resources Hub

Semester folders of study materials - admin uploads files or Drive links, students browse and open them.

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

Owner: Robi (student) + Fahmid (admin upload)

Key files: features/resources/*, features/admin/.../manage_resources_screen.dart

1. What it does

The Resources Hub is the department's shared file drawer. **Admins upload** any file (PDF/DOC/PPT/image/zip) or paste a Google Drive link, tagged to a semester and category. **Students browse** by semester folder -> category -> and tap to open. Files live in Supabase Storage; links open in the browser.

file	job
`resource_service.dart`	The only Supabase layer: fetch by semester, upload to the `resources` bucket, insert/delete rows.
`resource_controller.dart`	Student-side state: loads resources, filters by semester + category.
`resource_admin_controller.dart`	Admin-side: pick a file, upload, create the row, alert students.

2. Why resources filter by SEMESTER (not batch+section)

This is a deliberate contrast with routines. A 'Data Structures' note is useful to every student in that semester regardless of section, so resources key on **semester**. Routines key on batch+section because a schedule is cohort-specific. Good detail to mention - it shows you chose the data model per use-case.

3. Upload flow (admin)

1. Admin picks a file (file_picker, with bytes) or enters a Drive URL + sets semester + category.
2. `uploadFile(bytes, name, semester)` sanitises the filename, builds a path `<semester>/<timestamp>_<name>`, uploads to the public `resources` bucket, returns the public URL.
3. `createResource()` inserts the row, stamping `uploaded_by` from the session (required by the RLS insert policy).
4. A best-effort broadcast alerts students that new material was added.

Content-type is inferred from the extension so the browser opens the file correctly instead of downloading a blob.

4. Browse flow (student)

- `ResourceController` loads resources; the screen shows **semester folders** (all semesters - RLS lets students read every semester).
- Tapping a folder lists items; a category chip row filters client-side for instant response (no re-query).
- Tapping an item opens the URL via `url_launcher`, with a clipboard fallback if no browser can handle it.

5. Likely defense questions

question	answer
Where are the files stored?	Supabase Storage, the public `resources` bucket; the DB row stores the URL + metadata.
Why public bucket?	MVP simplicity; access is read-only browse. Could be tightened with signed URLs later.
How are uploads	RLS insert policy requires uploaded_by = the caller and an admin/teacher role.
Do students get notified?	Yes - upload fires a best-effort notification to students (in-app + push).